

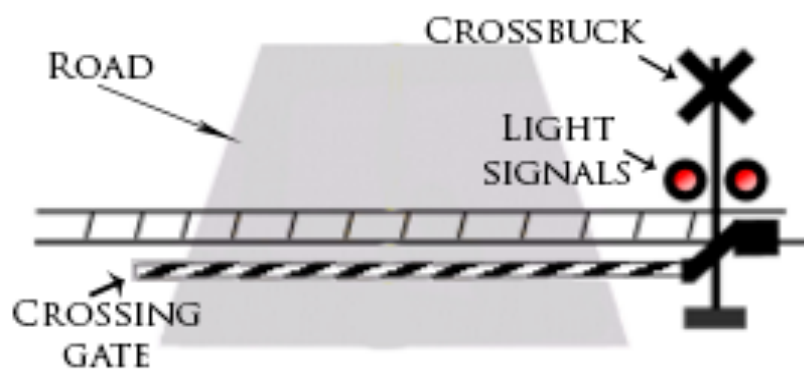
LEVEL CROSSING SYSTEM (SISTEMA DE PASSAGEM DE NÍVEL)

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STPA Step 1 – Define system purpose

1. Define and frame the problem

In the field of railway traffic control systems, level crossing (also known as grade crossing) is an intersection where a railway line crosses a road at same level. It is estimated that on average, each year 400 people in European Union and over 300 people in United States are killed in level crossing.



1.1 Considerations

- The roadway flows just one direction - one way road.
- An automated controller closes the crossing gate when the train is getting closer and opens after the train passed.
- A sensor is capable of identify if the train is getting closer and if it has passed.
- The crossing gate has light signals, but it does not have sound signals.
- Manual operation of the gate are not considered in this analysis.

1.2 Components

- Automated crossing gate controller
- 1 crossing gate
- 1 crossbuck
- 1 pair of light signals
- System sensor on the railway capable of identify whether the train is getting closer and whether it has passed.
- Railway
- Road

1.3 Stakeholders

- Train driver
- Vehicle driver
- Pedestrian
- Railway managers
- Railway company/authority

1.4 System purpose and goal

SG-1 - Avoid collision between train, cars, and pedestrians.

SG-2 - Provide safe passage for cars and pedestrians.

SG-3 - Do not interrupt train movement when approaching the level crossing.

SG-4 - Control traffic around level crossing system.

1.5 Identify losses and accidents

L1: Loss of human life or injury to humans

L2: Damage to vehicles or train

L3: Loss of reputation

L4: Economic loss

L5: Train derailment

1.6 Identify and map system hazards and system level constraints

1.6.1. Identify and map system hazards to losses

H1: Car/Pedestrian/animals on the railway [L-1] [L-2] [L-3] [L-4]

H1.1: Car crossing the railway

H1.2: Pedestrian crossing the railway

H1.3: Animals crossing the railway

H1.4: Car stopped on the railway

H1.5: Pedestrian stopped on the railway

H1.6: Animals stopped on the railway

H1.7: Some other physical obstacle on the railway

H2: Train approaching too fast [L-1] [L-2] [L-3] [L-4]

H3: Poor Signaling [L-1] [L-2] [L-3] [L-4]

H4: Vehicle/pedestrian trying to cross the railway when train is approaching and crossing gate is closing

1.6.2. Identify and map system safety constraints to system hazards

SC1: Car/Pedestrian MUST ONLY cross the railway when system signaling and crossing gate indicate it is safe crossing. [H1]
SC2: There must be no obstacles stopped on the railway [H1.1] [H1.2] [H1.3] [H1.4]
SC3: System must alert train approach [H1] [H3]
SC4: All signaling devices must be visible to drivers, pedestrians, and train driver [H1] [H3]
SC5: Vehicles must not pass through the crossing gate when it is being closed. [H-4]
SC6: The train must not be above a pre-determined speed [H2]

STPA Step 2 – Model the Control Structure